DIGITAL MEDIA FORENSIC INVESTIGATION

DIGITAL MEDIA FORENSIC REPORT

Forensic Media Investigation

CASE	**CASE-20260902-110111-797092**
EVIDENCE	**TEST-02-stego.bmp**
FILE TYPE	PC bitmap, Windows 3.x format, 1280 x 959 x 24, cbSize 3682614, bits offset 54
FILE SIZE	3.51 MB
SHA-256	`4d5a58294aec9faee3a8501dcfdf3cf975da0e9a2a3c1b72552750c9543e388c`

VERIFIED

Verified hidden-data recovery established

CORROBORATED_VERIFIED_HIDDEN_DATA	**LOW**	**HIGH**
FINAL ASSESSMENT	RISK	CONFIDENCE

The investigation established verified hidden-data recovery from the supplied evidence.

Recovered Content

```
This is a very secret message!
```

01 File Identification

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02 Triage

Printable strings identified	1788
Raw string output	Preserved in structured analysis artifacts; not reproduced in this report.

03 Metadata Forensics

Assessment	NO IMMEDIATE ANOMALY
GPS present	No
XMP present	No
Editing software detected	No
Comments present	No
Author present	No

04 Structure Analysis

Format	BMP
Valid signature	Yes
Trailing data	0 bytes
Embedded file signatures	None detected
Assessment	NO IMMEDIATE ANOMALY

05 Visual & Bit-Plane Analysis

Channel	Mean	Std. deviation	LSB one-ratio
Red	4.5992	14.5671	0.2180
Green	24.8528	56.1803	0.2550
Blue	16.4408	43.6391	0.2343

RGB channel intensity distributions differ The green channel has the highest mean intensity (24.8528), while the red channel has the lowest mean intensity (4.5992).

LSB distributions differ across RGB channels The green channel has the highest LSB one-ratio (0.2550), while the red channel has the lowest LSB one-ratio (0.2180).

Bit-plane entropy varies across channels and positions Measured bit-plane entropy varies across RGB channels and bit positions. These measurements describe the observed pixel distribution and are not, by themselves, evidence of hidden data.

06 Statistical Analysis

Channel	LSB one-ratio	Entropy
Red	0.2180	2.7307
Green	0.2550	3.9636
Blue	0.2343	3.8505

The green channel has the highest LSB one-ratio (0.255003), while the red channel has the lowest LSB one-ratio (0.217998).

The green channel has the highest measured entropy (3.963630), while the red channel has the lowest measured entropy (2.730689).

LSB one-ratios vary across image regions. The regional spread describes spatial variation in the measured LSB distribution and does not independently establish steganography.

Interpretation: Statistical and LSB measurements are supporting observations and are not independently treated as proof of hidden data.

07 Automated Steganalysis

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CANDIDATE OBSERVATIONS

Automated steganalysis results are treated as candidate observations. Confirmation is established separately through the verification stage.

08 Verification & Extraction

Verification status	VERIFIED
Technique	LSB steganography (b1,bgr,lsb,xy)
Verification score	0.950
Extraction successful	Yes
Recovered payload	/home/kali/steganalysis-framework/cases/CASE-20260902-110111-797092/extracted/zsteg_payload.txt

VERIFIED RECOVERY

The recovered artifact passed the Module 8 verification state.

Validated Recovered Content

```
This is a very secret message!
```

Steghide: Extraction was unsuccessful under the tested conditions; this does not establish absence of hidden data.

09 Payload Analysis

Payload status	IDENTIFIED

Classification	TEXT
File type	ASCII text, with no line terminators
Size	30 bytes
Payload SHA-256	da9c71e4083f422bcd16b7eac5d83138f5f28b39afc275fd1b8f4bf41a5f926f
Magic signature	Unknown / no known signature
Printable strings	1
YARA matches	0
Archive container	No

- No known magic-byte signature was identified.
- Payload is classified as text based on file-type analysis.
- 1 printable string(s) were identified.
- YARA scan completed with no matching rules.

10 YARA Detection

NO	0	0
YARA AVAILABLE	RULES LOADED	MATCHES

NO KNOWN YARA INDICATORS DETECTED

11 Evidence Correlation, Decision & Risk

2	1	YES
CORRELATION GROUPS	RISK SCORE	FURTHER ANALYSIS

2 corroboration group(s) contain multiple evidence items. The grouped findings should be evaluated together while preserving the original evidence states. Correlation alone does not establish maliciousness.

Decision

Final decision	CORROBORATED_VERIFIED_HIDDEN_DATA
Further analysis required	Yes

Decision rationale

- Verified forensic evidence is present and multiple related evidence items were correlated.
- The evidence supports a corroborated assessment of recovered hidden data.
- Verification of hidden data does not by itself establish that the recovered payload is malicious.
- Further payload analysis is required to characterize the recovered artifact and determine whether additional investigation is warranted.

Risk rationale

- 1 corroborated verified evidence group(s) contributed 1 additional point(s).
- The evidence produced an aggregate score of 1.

• The resulting evidentiary risk level is LOW.

• Verified hidden-data evidence carries greater weight because the underlying content was successfully recovered and validated.

12 Investigator Conclusion

• The investigation established verified hidden-data recovery with corroboration from multiple related evidence items.

• The evidentiary risk assessment was LOW with a score of 1.

• Further forensic analysis is recommended based on the existing evidence and decision-layer assessment.

• The findings describe the forensic evidence recovered during analysis and do not by themselves establish malicious intent.

Limitations

• Candidate evidence represents an investigative lead and should not be treated as verified hidden content.

• YARA matches are contextual indicators and do not automatically establish maliciousness.

• The reported risk score represents evidentiary suspicion and concealed-content significance; it is not a probability of maliciousness.

• The decision layer identified a need for further analysis of the available evidence or recovered artifacts.

Full machine-readable module results and raw tool artifacts are preserved separately. This PDF presents only selected evidence required for investigator interpretation and case-level reporting.